

Shape-Gate Loss Pilot Campaign Results Report

Shape-Gate Loss Pilot Closeout

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Executive Summary

The `shape_gate_loss_pilot_2026_07_20` campaign completed the approved one-run pilot on `polished_dataset` setpoints for the `Fw` surface.

The accepted campaign output directory is `output/training_campaigns/2026-07-20-19-59-14_shape_gate_loss_pilot_2026_07_20`.

The run completed without queue failures and produced the required campaign leaderboard, best-run YAML, best-run Markdown, metrics summary, training report, and checkpoint. It is accepted as a completed pilot, but it is not promoted to a full three-target, three-surface training campaign on scalar evidence alone.

Pilot result

- run: `te_shape_gate_loss_periodic_gru_sequence_fw__polished_setpoints` ;
- dataset: `polished_dataset` ;
- input mode: `setpoints` ;
- surface: `fw / forward_only` ;
- validation MAE: `0.002297 deg` ;
- validation RMSE: `0.002750 deg` ;
- test MAE: `0.002522 deg` ;
- test RMSE: `0.003133 deg` ;
- trainable parameters: `157,953` .

The scalar result is worse than the existing polished-setpoint `Fw` baselines in the reduced active set. The next decision should therefore be a strict `TE Curve Verification Pipeline` shape-gated evaluation of this checkpoint against the reduced selected-model set, not immediate expansion to the full matrix.

Campaign Scope

The pilot intentionally used the narrowest approved scope:

Field	Value
Campaign	<code>shape_gate_loss_pilot_2026_07_20</code>
Dataset	<code>polished_dataset</code>
Input mode	<code>setpoints</code>
Dataset schema	<code>polished_setpoint_curve_v1</code>
Surface	<code>Fw</code>
Base model	<code>periodic_gru_sequence</code>
Loss profile	<code>shape_gate_centered_offset_amplitude_harmonic</code>
Expected run count	<code>1</code>
Completed run count	<code>1</code>

Field	Value
Failed run count	0

The full promotion rule remains unchanged: any new model family that survives a pilot must later be trained across `simplified_setpoints` , `polished_setpoints` , and `polished_actual_values` , with separate `global` , `Fw` , and `Bw` surfaces.

Remote Execution

The first remote invocation failed during repository-root mapping because the default remote path pointed to the local Windows checkout path. The successful operator-equivalent command used the remote repository path on `xilab-remote` :

```
powershell .\scripts\campaigns\cross_wave\run_shape_gate_loss_pilot_campaign.ps1 -Remote -
RemoteRepositoryPath "C:\Users\Martina Salami\Documents\Davide\Physics-Informed-Neural-Networks" -
RemoteCondaEnvironmentName pinns_env
```

The local wrapper timed out before its final sync step, after the remote training had already launched. The completed queue entry and campaign artifacts were recovered manually from `xilab-remote` and synchronized into the local repository.

Artifact Audit

Artifact	Path
Campaign leaderboard	output/training_campaigns/2026-07-20-19-59-14_shape_gate_loss_pilot_2026_07_20/campaign_leaderboard.yaml
Campaign best run	output/training_campaigns/2026-07-20-19-59-14_shape_gate_loss_pilot_2026_07_20/campaign_best_run.yaml
Campaign best-run report	output/training_campaigns/2026-07-20-19-59-14_shape_gate_loss_pilot_2026_07_20/campaign_best_run.md
Training metrics	output/training_runs/shape_gate_loss_pilot_periodic_gru_sequence/2026-07-20-19-59-14_te_shape_gate_loss_periodic_gru_sequence_fw_polished_setpoints/metrics_summary.yaml
Training report	output/training_runs/shape_gate_loss_pilot_periodic_gru_sequence/2026-07-20-19-59-14_te_shape_gate_loss_periodic_gru_sequence_fw_polished_setpoints/training_test_report.md
Best checkpoint	output/training_runs/shape_gate_loss_pilot_periodic_gru_sequence/2026-07-20-19-59-14_te_shape_gate_loss_periodic_gru_sequence_fw_polished_setpoints/checkpoints/periodic_gru_sequence-epoch=007-val_mae=0.00229675.ckpt
Completed queue entry	config/training/queue/shape_gate_loss_pilot/completed/2026-07-20-19-59-14_001_001_shape_gate_loss_periodic_gru_sequence_fw.yaml

Metric Breakdown

Metric	Validation	Test
Total loss	0.008242	0.011919
MAE	0.002297	0.002522
RMSE	0.002750	0.003133
Pointwise loss	0.005697	0.008580
Centered curve-shape loss	0.004500	0.005242
Curve-offset loss	0.001197	0.003338
Curve-amplitude loss	0.015342	0.018935
Sparse harmonic-shape loss	0.000108	0.000113

Comparable Polished-Setpoint Baselines

The table compares the pilot only against local training runs that used `polished_setpoints`. These are scalar training metrics, not official curve-first verification results.

Family	Surface	Validation MAE	Test MAE	Decision
<code>periodic_mlp_harmonic</code>	<code>global</code>	0.001137	0.001270	Stronger scalar baseline
<code>periodic_mlp_harmonic</code>	<code>Fw</code>	0.001208	0.001442	Stronger scalar baseline
<code>wave4_1_mae_robust_loss</code>	<code>Fw</code>	0.001792	0.002109	Stronger scalar baseline
<code>periodic_gru_sequence</code>	<code>Fw</code>	0.001832	0.002108	Stronger same-backbone baseline
<code>wave4_2_quantile_p10_p50_p90</code>	<code>Fw</code>	0.001801	0.002141	Stronger scalar baseline
<code>shape_gate_loss_pilot_periodic_gru_sequence</code>	<code>Fw</code>	0.002297	0.002522	Completed pilot, not promoted

Acceptance Decision

The campaign is accepted as a completed pilot because:

- the one approved queue entry completed;
- the completed run uses `polished_dataset` setpoints and `Fw` as requested;
- the campaign leaderboard and best-run artifacts exist;
- the metrics summary, training report, checkpoint, and completed queue entry exist locally after manual remote sync;
- the active campaign state records the remote-path recovery and final result.

The model is not accepted as a new active leader because:

- test MAE is worse than the same-backbone `periodic_gru_sequence` `Fw` polished-setpoint baseline;
- test MAE is worse than the current reduced-set polished-setpoint `Fw` robust and quantile candidates;
- the run has not yet passed the calibrated shape-gated reranker;
- the pilot covers only one surface and one target, while promotion requires three targets and three surfaces.

Recommended Next Step

Do not launch the full Aries campaign yet.

The next step should be a checkpoint-level `TE Curve Verification Pipeline` adapter for this pilot run, followed by a calibrated shape-gated reranker pass on `polished_dataset` setpoints `FW`. If the pilot shows a material shape, harmonic, or phase improvement despite weaker scalar MAE, then prepare a second pilot with softened weights or a two-run weight ablation. If it fails the shape gate as well, stop this branch and keep the shape metrics as validation and selection tools rather than as a training objective.

Closeout Boundary

This closeout does not change the official `TE Curve Verification Pipeline` leaders and does not promote the shape-gate loss pilot to program baseline. The official selection boundary remains curve-first and per-surface.